

Blocking & confounding in 2^k Designs

Chapter 7

- **Blocking** is an experimental technique to control **Nuisance factors** during the experiment running
- The experiment should be run accordingly
- The idea of blocking was previously discussed, here we will focus on the application to 2^K designs
- We will consider
 - Designs with Replicates
 - Designs without Replicates

Blocking a replicated 2^k Factorial Design

- It is the same problem we discussed in chpt 5
- Let's assume to be able to run 2^k treatments in the **same block level** (material batch, shift, etc.)
- If we have n design replicates, each replicate is a level of the **blocking factor** (basically an additional factor with n levels)
- We can have **more than one** replicate in the same block
- As usual we should **randomize** the runs inside the block levels and, if possible, among the block levels (if the blocking factor is the day, we can not randomize among the block levels)

Let's consider the chemical process experiment previously described (Process yield, reactant and catalyst)

$K = 2$ Factors

$n = 3$ Replicates

Here is the usual sum of squares formula

Example

Table 7-1 Chemical Process Experiment in Three Blocks

	Block 1	Block 2	Block 3
	(1) = 28 $a = 36$ $b = 18$ $ab = 31$	(1) = 25 $a = 32$ $b = 19$ $ab = 30$	(1) = 27 $a = 32$ $b = 23$ $ab = 29$
Block totals:	$B_1 = 113$	$B_2 = 106$	$B_3 = 111$

$$y_{ijb} = \mu + \tau_i + \beta_j + \tau\beta_{ij} + Bl_b + \varepsilon$$

$$SS_{Blocks} = \sum_{b=1}^3 \frac{B_b^2}{4} - \frac{y_{...}^2}{12} = 6.50$$

2^K

$n2^K$

ANOVA Table

Table 7-2 Analysis of Variance for the Chemical Process Experiment in Three Blocks

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P -Value
→ Blocks	6.50	2	3.25		
A (concentration)	208.33	1	208.33	50.32	0.0004
B (catalyst)	75.00	1	75.00	18.12	0.0053
AB	8.33	1	8.33	2.01	0.2060
Error	24.84	→ 6	4.14		
Total	323.00	11 ←			

$$A * B \rightarrow (2-1)(3-1) = 2$$

$$B * B \rightarrow (2-1)(3-1) = 2 \quad \longrightarrow \quad df_E = 2+2+2=6$$

$$A * B * B \rightarrow (2-1)(2-1)(3-1) = 2$$

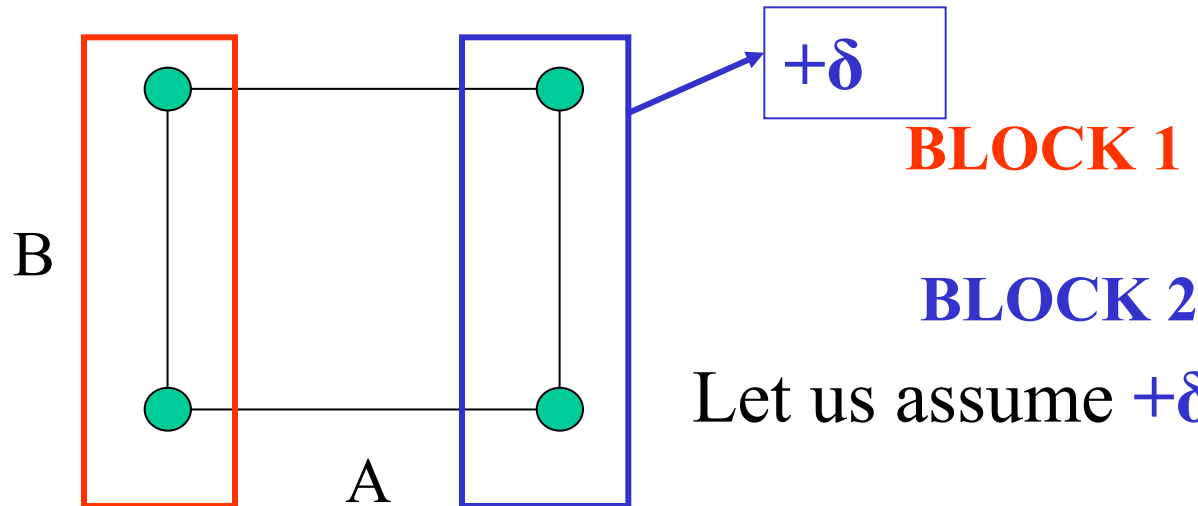
$$df_{TOT} = N - 1 = n2^2 - 1 = 3 * 4 - 1 = 11$$

“**Confounding**” in the 2^k Factorial Design

- Let us consider a design **without Replicates**. We can not use the previous technique.
- “**Confounding**” is a **design technique** for **arranging** a complete factorial design in blocks
- The technique implies that some information (usually high order interactions) will be **confounded with blocks** (e.g. **indistinguishable from blocks**)
- We will consider the case in which the number of Blocks is 2^p with $p < K$ (K = number of factors)
- i.e. with $K=6$ factors we want to arrange the 64 runs in 4 blocks ($p=2$) with 16 runs for each block

Confounding in two Blocks

- Let's consider a **single** replicate of a 2^2 design and suppose we are forced to run in two blocks

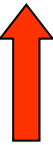


Let us assume $+\delta$ the Block Effect

$$EFF_A = \frac{1}{2}[-(1) + a + \delta - b + ab + \delta] = \frac{1}{2}[-(1) + a - b + ab] + \delta$$

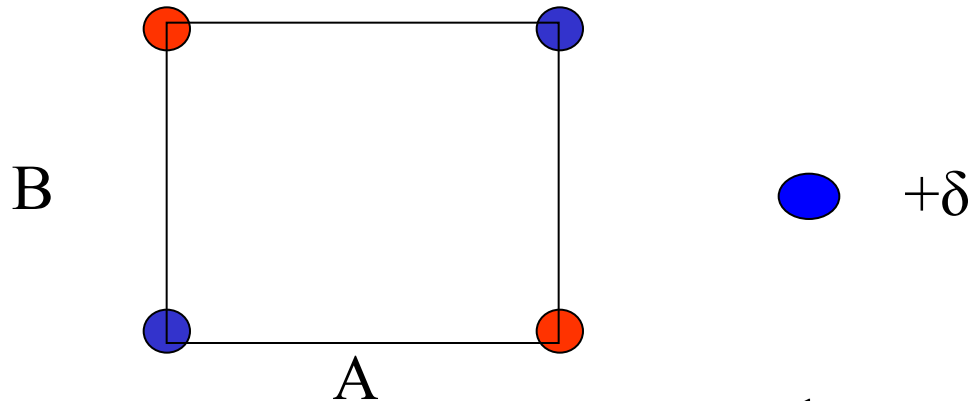
$$EFF_B = \frac{1}{2}[-(1) - a - \delta + b + ab + \delta] = \frac{1}{2}[-(1) - a + b + ab]$$

$$EFF_{AB} = \frac{1}{2}[+(1) - a - \delta - b + ab + \delta] = \frac{1}{2}[(1) - a - b + ab]$$



Could we do a better choice ?

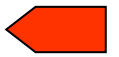
- A better choice is **confounding** the block effect with a less important model term
- We could confound the Block Effect with the Interaction



$$EFF_A = \frac{1}{2}[-(1) - \delta + a - b + ab + \delta] = \frac{1}{2}[-(1) + a - b + ab]$$

$$EFF_B = \frac{1}{2}[-(1) - \delta - a + b + ab + \delta] = \frac{1}{2}[-(1) - a + b + ab]$$

$$EFF_{AB} = \frac{1}{2}[+(1) + \delta - a - b + ab + \delta] = \frac{1}{2}[(1) - a - b + ab] + \delta$$

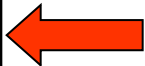
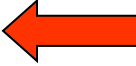


We can use the **Model (Sign) Table** for assigning the treatments to the Blocks

Block Equation
BI=AB

A	B	AB	
—	—	+	
+	—	—	
—	+	—	
+	+	+	

Confounding a 2^3 Design in two blocks

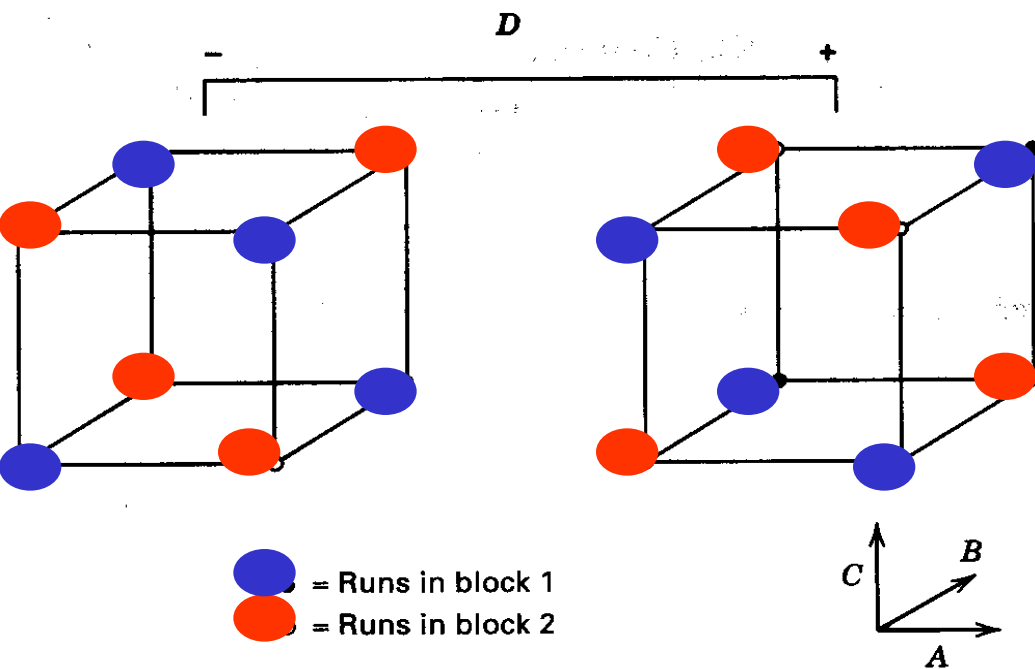
	A	B	C	AB	AC	BC	ABC	
(1)	—	—	—	+	+	+	—	
a	+	—	—	—	—	+	+	
b	—	+	—	—	+	—	+	
ab	+	+	—	+	—	—	—	
c	—	—	+	+	—	—	+	
ac	+	—	+	—	+	—	—	
bc	—	+	+	—	—	+	—	
abc	+	+	+	+	+	+	+	

Example 6-2 Single replicate 2^4 Factorial Design

Table 6-10 Pilot Plant Filtration Rate Experiment

Ordine std	Factor				Run Label	Filtration Rate (gal/h)
	A	B	C	D		
1	—	—	—	—	(1)	45
2	+	—	—	—	<i>a</i>	71
3	—	+	—	—	<i>b</i>	48
4	+	+	—	—	<i>ab</i>	65
5	—	—	+	—	<i>c</i>	68
6	+	—	+	—	<i>ac</i>	60
7	—	+	+	—	<i>bc</i>	80
8	+	+	+	—	<i>abc</i>	65
9	—	—	—	+	<i>d</i>	43
10	+	—	—	+	<i>ad</i>	100
11	—	+	—	+	<i>bd</i>	45
12	+	+	—	+	<i>abd</i>	104
13	—	—	+	+	<i>cd</i>	75
14	+	—	+	+	<i>acd</i>	86
15	—	+	+	+	<i>bcd</i>	70
16	+	+	+	+	<i>abcd</i>	96

Suppose we have to carry out $2^3=8$ runs per block



(a) Geometric view

Block 1	Block 2
(1) = 25	$a = 71$
$ab = 45$	$b = 48$
$ac = 40$	$c = 68$
$bc = 60$	$d = 43$
$ad = 80$	$abc = 65$
$bd = 25$	$bcd = 70$
$cd = 55$	$acd = 86$
$abcd = 76$	$abd = 104$

(b) Assignment of the 16 runs to two blocks

Our choice is
BI=ABCD

To simulate the Block effect, the observations in block 1 have been reduced by 20

Figure 7-4 The 2^4 design in two blocks for Example 7-2.

Effect Estimates

Table 7-6 Effect Estimates for the Blocked 2⁴ Design in Example 7-2

Model Term	Regression Coefficient	Effect Estimate	Sum of Squares	Percent Contribution
<i>A</i>	10.81	21.625	1870.5625	26.30
<i>B</i>	1.56	3.125	39.0625	0.55
<i>C</i>	4.94	9.875	390.0625	5.49
<i>D</i>	7.31	14.625	855.5625	12.03
<i>AB</i>	0.062	0.125	0.0625	<0.01
<i>AC</i>	-9.06	-18.125	1314.0625	18.48
<i>AD</i>	8.31	16.625	1105.5625	15.55
<i>BC</i>	1.19	2.375	22.5625	0.32
<i>BD</i>	-0.19	-0.375	0.5625	<0.01
<i>CD</i>	-0.56	-1.125	5.0625	0.07
<i>ABC</i>	0.94	1.875	14.0625	0.20
<i>ABD</i>	2.06	4.125	68.0625	0.96
<i>ACD</i>	-0.81	-1.625	10.5625	0.15
<i>BCD</i>	-1.31	-2.625	27.5625	0.39
Blocks (<i>ABCD</i>)		-18.625	1387.5625	19.51

same estimates

-18.625 = 1.375 (previous ABCD) - 20 (Block Effect)

$$SS_{BL} = \sum_{i=1,2} \frac{B_i^2}{2^{K-1}} - \frac{y_{\dots}^2}{N}$$

ANOVA Table

Table 7-7 Analysis of Variance for Example 7-2

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P -Value
Blocks (<i>ABCD</i>)	1387.5625	1			
<i>A</i>	1870.5625	1	1870.5625	89.76	<0.0001
<i>C</i>	390.0625	1	390.0625	18.72	0.0019
<i>D</i>	855.5625	1	855.5625	41.05	0.0001
<i>AC</i>	1314.0625	1	1314.0625	63.05	<0.0001
<i>AD</i>	1105.5625	1	1105.5625	53.05	<0.0001
Error	187.5625	9	20.8403		
Total	7111.4375	15			

The *ABCD* interaction (confounded with Block Effect) is **not** considered in the error term

The remaining rows are not changed

More than two blocks

- Let's consider the 2^3 Factorial Design
- The solution we just found in 2^3 exp is expressed by the following Block Equation:

$$Bl_1 = ABC$$

- If we want to arrange the experimental runs in **four blocks** we need a **second** Block Equation.
i.e.

$$Bl_2 = BC$$

- Using the two Block Equations we can find the other confounded effects

$$Bl_1 = ABC$$

$$Bl_2 = BC$$

and $Bl_1 Bl_2 = ABC * BC = A$

Summing up our choice **confounds** the main effect A , ABC and BC

Please VERIFY with a simulated numerical example

- A better choice is to select

$$\mathbf{B}\mathbf{l}_1 = \mathbf{A}\mathbf{B}$$

$$\mathbf{B}\mathbf{l}_2 = \mathbf{A}\mathbf{C}$$

- And we will confound

$$\mathbf{B}_1\mathbf{B}_2 = \mathbf{A}\mathbf{B} * \mathbf{A}\mathbf{C} = \mathbf{B}\mathbf{C}$$

- **But** Main effects will **not** be confounded
- To find the right solution we can use Table 7.8 (textbook) or JMP sw

Table 7-8 Suggested Blocking Arrangements for the 2^k Factorial Design

Number of Factors, k	Number of Blocks, 2^p	Block Size, 2^{k-p}	Effects Chosen to Generate the Blocks	Interactions Confounded with Blocks
3	2	4	<i>ABC</i>	<i>ABC</i>
	4	2	<i>AB, AC</i>	<i>AB, AC, BC</i>
4	2	8	<i>ABCD</i>	<i>ABCD</i>
	4	4	<i>ABC, ACD</i>	<i>ABC, ACD, BD</i>
	8	2	<i>AB, BC, CD</i>	<i>AB, BC, CD, AC, BD, AD, ABCD</i>
5	2	16	<i>ABCDE</i>	<i>ABCDE</i>
	4	8	<i>ABC, CDE</i>	<i>ABC, CDE, ABDE</i>
	8	4	<i>ABE, BCE, CDE</i>	<i>ABE, BCE, CDE, AC, ABCD, BD, ADE</i>
	16	2	<i>AB, AC, CD, DE</i>	All two- and four-factor interactions (15 effects)
6	2	32	<i>ABCDEF</i>	<i>ABCDEF</i>
	4	16	<i>ABCF, CDEF</i>	<i>ABCF, CDEF, ABDE</i>
	8	8	<i>ABEF, ABCD, ACE</i>	<i>ABEF, ABCD, ACE, BCF, BDE, CDEF, ADF</i>
	16	4	<i>ABF, ACF, BDF, DEF</i>	<i>ABE, ACF, BDF, DEF, BC, ABCD, ABDE, AD, ACDE, CE, BDF, BCDEF, ABCEF, AEF, BE</i>
	32	2	<i>AB, BC, CD, DE, EF</i>	All two-, four-, and six-factor interactions (31 effects)
7	2	64	<i>ABCDEFG</i>	<i>ABCDEFG</i>
	4	32	<i>ABCFG, CDEFG</i>	<i>ABCFG, CDEFG, ABDE</i>
	8	16	<i>ABC, DEF, AFG</i>	<i>ABC, DEF, AFG, ABCDEF, BCFG, ADEG, BCDEG</i>
	16	8	<i>ABCD, EFG, CDE, ADG</i>	<i>ABCD, EFG, CDE, ADG, ABCDEFG, ABE, BCG, CDFG, ADEF, ACEG, ABFG, BCEF, BDEG, ACF, BDF</i>
	32	4	<i>ABG, BCG, CDG, DEG, EFG</i>	<i>ABG, BCG, CDG, DEG, EFG, AC, BD, CE, DF, AE, BE, ABCD, ABDE, ABEF, BCDE, BCEF, CDEF, ABCDEFG, ADG, ACDEG, ACEFG, ABDFG, ABCEG, BEG, BDEFG, CFG, ADEF, ACDF, ABCF, AFG</i>
	64	2	<i>AB, BC, CD, DE, EF, FG</i>	All two-, four-, and six-factor interactions (63 effects)

Statistical Analysis

(for Full and Balanced Experiment)

- Compute the SS for each **Effect**
- SS_{BLOCK} is the **sum** the confounded effect SS s
- df_{BLOCK} is the **number** of confounded effects

Example

- We want to design a 2^6 (**K=6**) factorial design with 8 blocks (**p=3**),
- Each Block will have $8 = 2^{K-p}$ treatment conditions
- Table 7.8 suggests the following Block Equations

$$B1_1 = AB EF$$

$$B1_2 = ABCD$$

$$B1_3 = ACE$$

Please manually find the confounded effects and check using the table

Partial Confounding

- If we have few df_E for MS_E we can replicate the experiment (**k=3 p=1 n=4 N=32**)

Replicates	1		2		3		4	
Blocks	1	2	3	4	5	6	7	8
	(1)	abc	(1)	abc	(1)	abc	(1)	abc
	ac	a	ac	a	ac	a	ac	a
	ab	b	ab	b	ab	b	ab	b
	bc	c	bc	c	bc	c	bc	c

Remark: Interaction **ABC** remains confounded

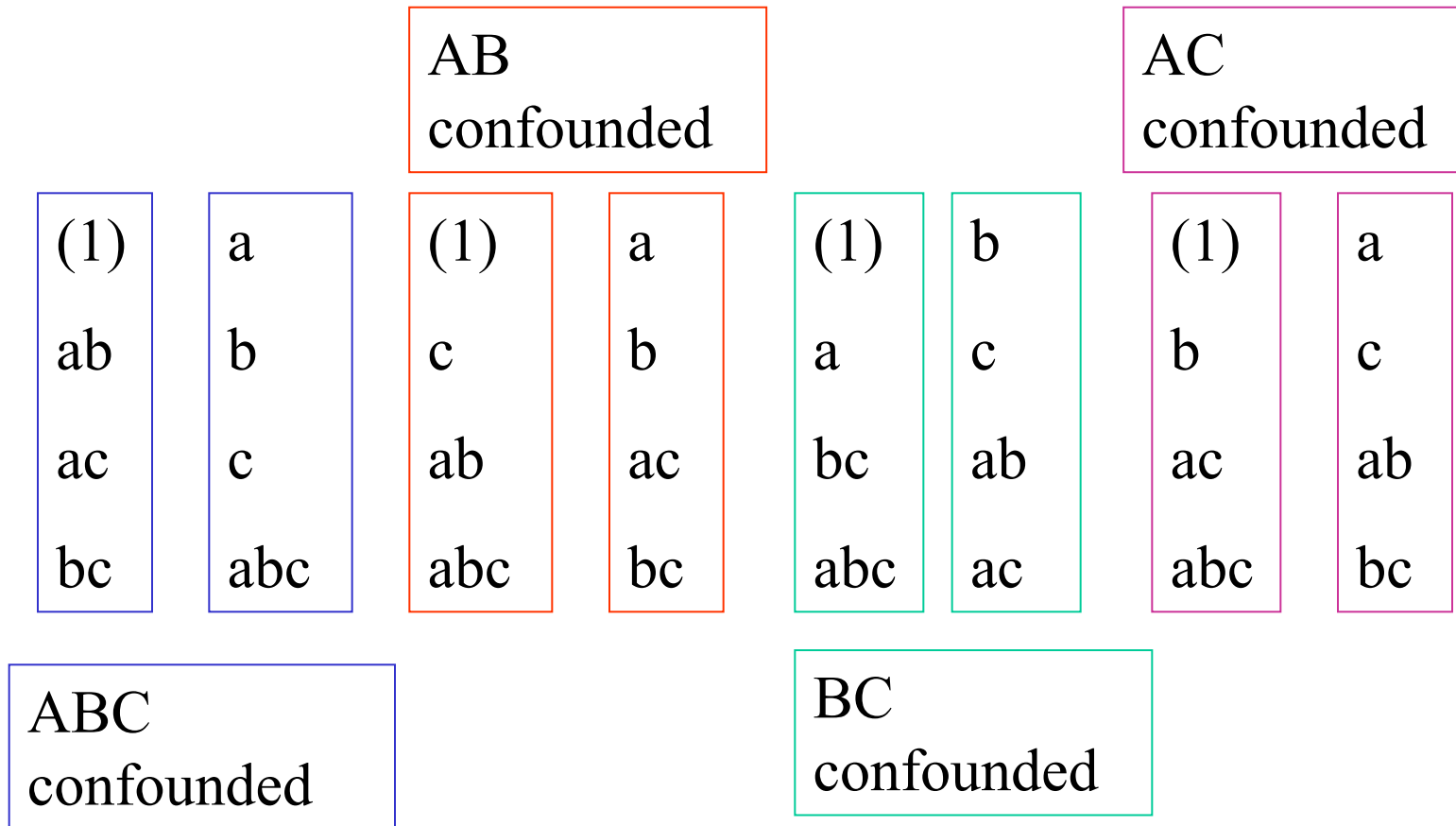
ANOVA Table

<u>Source</u>	<u>df</u>
Blocks	(8-1)=7
A	1
B	1
C	1
AB	1
AC	1
BC	1
<u>Error</u>	<u>18</u>
Total (corr)	31

$$SS_{Blocco} = \sum_{l=1, n2^p} \frac{B_l^2}{2^{K-p}} - \frac{y_{...}^2}{n2^k} =$$

$$= \sum_{l=1,8} \frac{B_l^2}{4} - \frac{y_{...}^2}{n2^k}$$

Partial confounding is an experimentation technique that assigns the runs differently: in each **replicate** a different effect is confounded as follows:



ANOVA Table

<u>Source</u>	<u><i>df</i></u>
Block	7
A	1
B	1
C	1
AB (from replicate 1,3,4)	1
AC (from replicate 1,2,3)	1
BC (from replicate 1,2,4)	1
ABC (from replicate 2,3,4)	1
<u>Error</u>	<u>17</u>
Total (Corr)	31

← One *df* less
because we
estimate ABC

Example

- Consider the example of the **bottling line** (Factors: A-%CO2, B-pressure, C-line rate; Response: Deviation of the filling height from the target).
- 4 treatments per block and two replicates

ABC
confounded

(1) = -3

ab = 2

ac = 2

bc = 1

a = 0

b = -1

c = -1

abc = 6

AB confounded

(1) = -1

c = 0

ab = 3

abc = 5

a = 1

b = 0

ac = 1

bc = 1

$$SS_{ABC} = \frac{CONT_{ABC}^2}{n2^k} = \frac{[a + b + c + abc - ab - ac - bc - (1)]^2}{1 * 2^3} = \frac{[1 + 0 + 0 + 5 - 3 - 1 - 1 + 1]^2}{8} = 0.5$$

$$SS_{AB} = \frac{CONT_{AB}^2}{n2^k} = \frac{[(1) + abc - ac + c - a - b + ab - bc]^2}{1 * 2^3} = 0.5$$

$$SS_{BLOCK} = \sum_{l=1,4} \frac{B_l^2}{4} - \frac{y_{...}^2}{16} = \frac{2^2 + 4^2 + 7^2 + 3^2}{4} - \frac{16^2}{16} = 3,5$$

$$SS_{TOT} = \sum_{i,j,k,l} y_{ijkl}^2 - \frac{y_{...}^2}{n2^K} = 94 - \frac{16^2}{16} = 78$$

$$SS_A = \frac{CONT_A^2}{n2^K} = \frac{[1+5+3+11-(-4)-(-1)-(-1)-2]^2}{\textcolor{red}{2} * 2^3} = \frac{24^2}{16} = 36$$

$$SS_B \dots SS_C \dots SS_{AC} \dots SS_{BC}$$

$$SS_E = SS_{TOT} - SS_{BLOCK} - SS_{[]} - \dots - SS_{[]}$$

Source	SS	<i>df</i>	MS	F	p-value	ANOVA Table
Block	3.50	3	1.16			
A (CO2%)	36.0	1	36.0	48.0	0.0001	←
B (pressure)	20.25	1	20.25	27.0	0.0035	←
C (Speed)	12.25	1	12.25	16.33	0.0099	←
AB(rep1)	0.5	1	0.5	0.67	0.45	
AC	0.25	1	0.25	0.33	>0.5	
BC	1.00	1	1.00	1.33		
ABC(rep2)	0.50	1	0.50	0.67	0.45	
Error	3.75	5	0.75			
Total (corr)	78.00	15				

Confounding summary

- **Without replicates**
 - 2^K Factorial Designs can be confounded in 2, 4, 8, ... (2^p , $p > 1$) Blocks
 - For 2^p Blocks, you need **p Block Equations**
 - The **Confounded Effects** are found by all possible Block Equation multiplications ($2^p - 1$)
 - The choice of the Block Equations is not trivial. Use the Table 7-8 or Sw suggestions
- **Partial Confounding with replicates**